



Signaling Physical Exertion. Can Observers Detect Physical Exertion from Athletes' Nonverbal Behavior

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Abstract

This research examined whether observers can accurately detect physical exertion and how their ratings align with athletes' self-reports and physiological indicators. Across five independent studies (conducted 2024–2025 in Germany), 276 observers evaluated exertion from short video clips of 103 athletes differing in age (children and adults), exertion stage (rested, medium, high), and activity type (sprinting, running, cycling, workouts, track and field) to test the primary hypothesis that people can detect exertion based on thin slices of behavior. Observers reliably detected exertion based on the results of ANOVA and Pearson correlation analyses: ratings increased with physical load and correlated strongly with both self-reports and heart rate data, indicating reliance on valid perceptual cues. Comparisons showed slightly stronger correlations between self-reports and heart rate than between observer ratings and heart rate. Notably, after highly strenuous tasks, observers tended to underestimate exertion relative to athletes' self-ratings. These findings highlight the accuracy of thin-slice judgments of exertion while revealing systematic underestimation at peak effort, offering insights with practical implications for sports performance monitoring and training feedback.

Keywords Person perception · Social cognition · Nonverbal behavior/Communication; · Facial expressions · Body language · Sports · Fatigue

Introduction

Feeling states such as physical exertion or physical fatigue arise from sensing and integrating diverse bodily and contextual cues. Importantly for the present research, some of these cues are externally observable, enabling perceivers to infer others' feeling states through expressive behavior. However, little is known about how perceptions of one's own exertion—based on interoceptive signals (Craig, 2002)—correspond to perceptions of others' exertion, which rely on interpreting bodily cues. A recent review (Liu et al., 2025) high-

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lighted the scarcity of systematic research on how people judge exertion in others and how these judgments relate to felt exertion. The current work addresses this gap by comparing individuals' own ratings of perceived exertion (Borg, 1962) with observers' ratings and physiological indicators across varying levels of exertion in different exercises. Our primary aim is to test whether observers can accurately detect exertion in others and how their judgments align with self-ratings and physiological signals.

Understanding what someone else feels by interpreting facial, bodily, and behavioral cues is fundamental to social interaction and is typically discussed under the umbrella of empathy. Darwin (1871) already proposed that humans evolved sensitivity to others' internal states, noting that observing another's fatigue can evoke recollections of one's own. This idea aligns with the concept of *retrodiction*, a process by which a perceiver infers a target's internal mental or physical state by reasoning backward from observable behavior (Gallese & Goldman, 1998). In this framework, the observer uses the athlete's external cues as a basis to reconstruct the likely subjective experience that preceded or accompanied the action. Although the term *retrodiction* is relatively recent, the underlying process—inferring others' internal states from expressive cues—has long been examined across scientific disciplines. *Retrodiction* is a useful framework for understanding how people infer exertion from observed behavior. Research on facial expressions shows that expressions reveal emotional states (Zaki et al., 2008, 2009), which themselves reflect external events (Cassidy et al., 2015; Kang et al., 2017; Pillai et al., 2012, 2014; Sheppard et al., 2016; Teoh et al., 2017; Valanides et al., 2017; Wu et al., 2016). Thus, internal states are at least partially visible through behavior (Gallagher, 2011), and accurate state inference depends on interpreting these cues.

A key aspect of *retrodiction* is that perceivers must infer not only inner states but also the prior event that caused them (Teoh et al., 2017). Inferring unseen causes from observed behavior therefore requires integrating knowledge about both the person and the situational context. While expressive behavior does not map perfectly onto specific events, many spontaneous reactions fall within recognizable ranges (e.g., certain bodily and facial configurations as a response to victory and defeat; Tracy & Matsumoto, 2008), allowing perceivers to make reasonably accurate inferences from subtle behavioral differences (Furley, 2023).

The thin-slices paradigm (Ambady & Rosenthal, 1992; Ambady et al., 2000; Carney et al., 2007) offers compelling evidence (i.e., consistency across contexts and meta-analytic support) that people can make accurate judgments based on brief observations of expressive behavior. In this paradigm, participants view short video segments or photographs and judge attributes of the target. Research consistently shows that individuals can infer a wide range of traits and states from minimal nonverbal cues (Murphy & Hall, 2021, for a recent review). This paradigm has recently been extended to sport. Observers viewing short clips of athletes' nonverbal behavior can distinguish whether athletes are leading or trailing during competitions (e.g., table tennis, basketball, soccer, handball; Furley & Schweizer, 2014) and judge leading athletes as more confident, proud, and dominant (2016a; Furley et al., 2023). Similarly, observers can infer whether soccer referees are communicating decisions about ambiguous versus unambiguous fouls. Referees appear less confident when signaling decisions on ambiguous situations, even though observers see only thin slices of behavior and not the decision context itself (Furley & Schweizer, 2016b).

Building on this work, the present research examines *retrodiction* in the domain of physical exercise by testing whether observers differentiate perceived exertion across groups

exposed to varying exertion levels and how observer ratings correspond to self-ratings and physiological measures. Borg (1962) conceptualized perceived exertion as a “Gestalt” integrating sensations from muscles, skin, and joints, along with perceptions of fatigue, strain, heat, pain, or anxiety. This led to the development of the Borg Rating of Perceived Exertion (RPE) Scale, which ranges from 6 to 20 and correlates roughly with heart rate (Borg, 1998).

Previous research shows that nonverbal expressions during physical tasks influence observers’ judgments of effort (Rejeski & Lowe, 1980)—a theoretical construct that is arguably linked to exertion. Facial expressions, particularly activity of frowning-related muscles, increase with exertion (De Morree & Marcora, 2010, 2012; Uchida et al., 2018). Additional observable cues include reddened skin, sweating, labored breathing, and visible blood-vessel dilation (Borg, 1998). These cues should support perceivers in inferring exertion, although some of these cues like reddened skin may differ in visibility depending on skin color. Yet, as Liu et al. (2025) noted in their review “Perceiving exertion in others: From interoception to exteroception,” systematic research on how observers detect exertion remains limited. Only ten studies met inclusion criteria, suggesting piecemeal evidence that observers can detect exertion but leaving major methodological gaps. To address these shortcomings, our study applies the thin-slices paradigm to test whether perceivers (a) detect physical exertion from brief observations of nonverbal behavior, (b) show correspondence between their ratings and targets’ self-reported exertion, and (c) align with physiological measures. This research is important because accurate assessment of others’ exertion is crucial in sport, healthcare, and occupational settings, where coaches, therapists, and employers rely on such judgments to monitor performance and manage workload. Despite this relevance, little is known about how people perceive exertion in others or how these perceptions relate to underlying physiology. As Liu et al., (2025, p. 2710) conclude: “This is still a wide-open field, ripe for further study.”

The Present Research

In the present research, we recorded individuals before, during, and after various physical activities and asked independent observers to rate the perceived exertion displayed in these video clips. We then compared aggregated observer ratings with participants’ self-reported RPE scores and their heart rate responses. Recent work (Furley et al., 2025) demonstrates that aggregated judgments—commonly referred to as the “wisdom of the crowd”—can be highly accurate because averaging across diverse and independent observers cancels out individual noise. For this mechanism to yield accurate assessments, however, valid cues of exertion must be visible to observers. Accordingly, accuracy in the present studies was defined as the correspondence between observer ratings, self-reported exertion, and heart rate.

Based on prior findings from the thin-slices paradigm and theorizing on collective judgment accuracy (Furley et al., 2025), we hypothesized that observers would reliably detect physical exertion from brief video segments—i.e., that averaged observer ratings would significantly differ as a function of prior physical exertion. Drawing upon paradigms established in seminal personality research, the current study extends the comparison of self- and other-reports—traditionally used to assess stable traits such as personality and intelligence (Borkenau et al., 2004; Connelly & Ones, 2010)—to the transient state of physical exertion. Central to this approach is the established finding that observer ratings demonstrate

comparable reliability to self-reports, effectively challenging the assumption that external judgments are inherently ‘noisier’ or less valid than first-person accounts. More specifically, across five studies we predicted that (a) participants’ own RPE ratings would correlate with aggregated observer ratings (calculated as the mean of all observer ratings in the respective situation), and (b) both subjective measures—self and observer—would correlate with the physiological indicator heart rate. To establish the robustness of the primary hypothesis—that observers can accurately infer exertion—a multi-method approach involving five different exertion protocols was employed. This design has the potential to provide converging evidence across diverse physical tasks. However, it should be noted that these procedures were not preregistered, and the analyses regarding specific exertion types and population differences remained exploratory.

Overview of Studies

Across the five studies, observers judged exertion from short pre-, (mid)-, or post-exercise clips from different athlete groups: Study 1: Male and female 200-m sprinters, pre- and post-race; Study 2: Male and female 3000-m runners, pre- and post-race; Study 3: Male and female cyclists on a home trainer, at pre-, mid-, and full-exertion stages; Study 4: Male and female recreational athletes performing four sets of 10 burpees, rated before the first set, after the first set, and after the fourth set; Study 5: Male and female child track athletes (6–8 years old), before and after a 400-m run. The chosen physical activities across the 5 studies were intended to represent a broad spectrum of common exerting activities. Across these studies, we tested whether observers could infer exertion levels from thin slices of behavior, and whether these judgments aligned with self-reported exertion and physiological load. We used the mean of observers’ RPE as an aggregated measure of “other’s RPE”. Across all five studies, intraclass correlation coefficients for the observer ratings of athletes’ exertion were good to excellent (ranging from .82 to .98) justifying the use of the mean other ratings.

Five samples of observers estimated the exertion of the athletes in the present research. Sample size considerations were based on previous thin-slice research in sports (Furley et al., 2025; Furley & Schweizer, 2014) and similar to thin-slices studies in the meta-analysis of Ambady and Rosenthal (1992). As there is no straight forward method of computing power in a full 2 x 2 or 2 x 3 within-subject ANOVA we only made sure that sample size was adequate (i.e., sensitive) to detect small effects in a 1 x 2 or 1 x 3 within-subject design ANOVA to determine the main research question if observers could detect physical exertion from brief video segments. All samples of participants did not have any particular domain-specific knowledge in sports or track and field. More specifically, it was ensured that the participants who observed and rated the exertion in the videos were no experts in the observed sport (e.g., a track-and-field coach judging exertion in the sprint or 3000m race) as it has been suggested that exercise experience may enhance sensitivity to others’ exertion (e.g., Gonzales et al., 2015). The study protocols received ethical approval of the university’s ethics board. This study was not preregistered. All data can be found here: https://osf.io/w4mk/y/?view_only=902e13789fb849f483800fca6923f27c.

Study 1: Adult 200m Sprint

Method

Participants

The observer sample estimated the exertion of 21 different track and field athletes, six of whom were female and fifteen male. The average age of the athletes was 24 years. All recruited athletes completed the study. The observer sample consisted of 53 participants (26 female and 27 male) with an average age of 29.45 years ($SD = 6.23$). Only participants who completed the online survey were included in the analyses.

Written and digital informed consent was obtained from all athletes and observers. The protocol involved two recording conditions (see next section for further details): an overt condition, where participants were instructed to stand before a camera, and a covert condition, where participants were filmed via a laptop webcam while recording their RPE scores. This deceptive element was necessary to capture naturalistic behavior; consequently, all participants were fully debriefed regarding the covert recording upon completion of the study. The experimental protocol received formal approval from the university's ethics committee.

Procedure and Stimuli Creation

Each sprinter was filmed at two time points on an outdoor track and field training facility: before exertion, and immediately following a 200-meter sprint at maximum intensity (within seconds after crossing the finish line). Heart rate was measured via pulse oximeter (Polar Pacer Pro). After each recording, athletes were asked to rate their perceived exertion using the Borg Scale (ranging from 6 to 20) and enter their rating into an Excel spreadsheet on a laptop. While they were entering their ratings, they were filmed again with the laptop webcam without their knowledge. These unobtrusively filmed clips served the purpose to test if athletes changed their demeanour if they knew that they were being filmed. As ratings of unobtrusively filmed clips and overtly filmed clips were rated almost identically (no difference in dependent t-test $p = .75$), only the overtly filmed clips were used in further analyses. All videos were recorded without audio to ensure evaluations were based solely on visual cues. The video sequences ranged in length from three to six seconds and show only the face and parts of the upper body.

For the assessment of the observer RPE ratings an online survey was conducted in which the observer sample rated perceived exertion in the video stimuli on the Borg RPE Scale ranging from 6 (no exertion) to 20 (maximal exertion). We used the software Unipark to administer the online study. The software displayed the respective samples of stimuli in different random orders for every participant. This approach helps to ensure that results do not depend on specific sequences of stimuli. After every video clip, participants were asked to estimate the respective perceived exertion on the BORG RPE scale of the athlete they had just seen. All videos were presented silently to ensure that ratings were based on nonverbal behavior. Each clip was embedded within the survey interface, with instructions displayed above the video. Participants provided their ratings using a digital slider corresponding to the Borg RPE Scale (6–20), which was positioned directly beneath the video. After giving

their rating, participants proceeded to the next clip by clicking “Next.” Backtracking was disabled in the software.

Before commencing the experiment, participants filled out a questionnaire gathering demographic data (age and gender). After completing the testing procedure, participants were informed about the purpose of the experiment.

Statistical Analyses

In line with the rationale of this work, the statistical analysis aimed to test if observers of short recordings of different groups of athletes in different contexts can accurately infer exertion of these athletes. To this end two main analysis approaches were chosen (for all five studies). First, we conducted separate 2 (type of rating: own vs. averaged other rating) $\times$ 2 (or 3 in studies 4 and 5) (level of exertion: no exertion vs. [medium] vs. high exertion within-subject ANOVAs depending on the stimulus set. We follow up significant interaction effects with post-hoc simple effect analyses. Second, we report Pearson correlation analyses (and 95% confidence intervals) quantifying the relationship between own, other RPE scores, and physiological heart rate data. We denote results as significant if p -values fall below 0.01. Statistical analyses were performed using IBM SPSS Statistics (Version 31). Prior to the primary analyses, all data were screened to ensure they met the necessary assumptions for the respective statistical tests.

Results

ANOVA

The descriptive statistics are summarized in Fig. 1, left panel. A 2 (own vs. observer rating) $\times$ 2 (before vs. after exertion) ANOVA (see left panel of Fig. 1) yielded a significant main effect of rating type, $F(1, 20) = 13.31, p = .002, \eta^2_p = .40$, a significant main effect of exertion, $F(1, 40) = 364.37, p < .001, \eta^2_p = .95$, and a significant interaction, $F(2, 40) = 24.39, p < .001, \eta^2_p = .55$. In line with expectations, exertion ratings were higher post-sprint than at baseline for both self-ratings and observer-ratings (both $p < .001$). Follow-up simple effects analyses further clarified the significant interaction: observers’ ratings did not differ significantly from athletes’ RPEs under rested conditions ($p = .117$), but observers significantly underestimated exertion relative to the athletes’ own ratings following the sprint ($p < .001$). The main effect of rating type reflects consistently higher self-ratings than observer ratings overall.

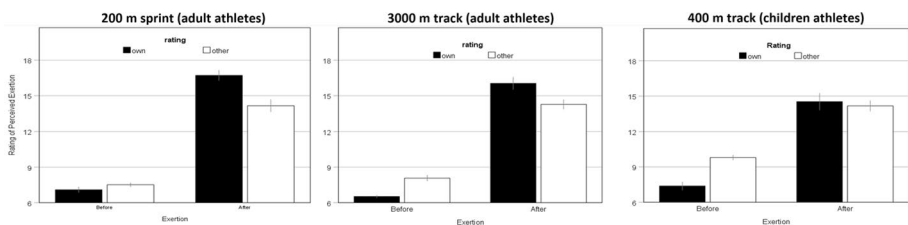


Fig. 1 Summary of descriptive results from Study 1-3. Error bars represent standard errors

Correlational Analyses

Figure 2, left panel illustrates the strong positive correlation between own and averaged observer RPE, $r(42) = .893, p < .001$. Heart rate data for the 200-m adult sprint were unavailable due to a measurement device malfunction in that study.

Discussion

The findings demonstrate high convergence between self-reports and observer judgments, evidenced by the strong positive correlation between athletes' RPE and aggregated thin-slice ratings. While both athletes and observers successfully tracked the increase in exertion, a significant interaction revealed that observers' accuracy diverged at higher intensities. Specifically, observers accurately assessed baseline states but systematically underestimated exertion following the sprint. This suggests that while external raters can reliably rank-order relative effort, they may lack sensitivity to the peak physiological strain experienced by the athlete.

To test the generalizability of our findings, Study 2 extended the logic of Study 1 to the context of an adult 3,000-meter track race. This approach follows Fiedler's (2011) recommendation to replicate effects using diverse stimuli, ensuring the observed phenomenon is robust and not artifactual to a specific set of stimulus material. By varying the athletic domain, we aim to confirm that the convergence between self- and observer-reports generally applies to the perception of exertion.

Study 2: Adult 3000m Track

Method

Participants

The observer sample estimated the exertion of 21 different track and field athletes (11 male, 10 female). The average age of the athletes was 23 years. All recruited athletes completed the study. The observer sample of 53 participants (28 female and 25 male) with an average age of 24.42 years ($SD = 8.01$). Only participants with complete data in the online survey were included. Sample size considerations and consent procedure were identical to Study 1.

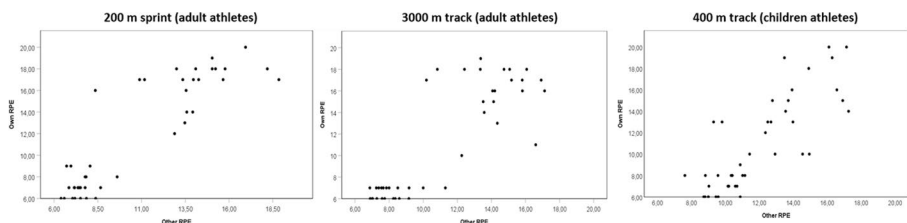


Fig. 2 Scatter plots illustrating the relationship between own ratings and observer ratings of RPE for Studies 1–3

Procedure and Stimuli Creation

The procedure and video recordings were highly similar to Study 1 and therefore only the important differences are described here. The stimuli comprised 42 video sequences, each lasting two to four seconds, featuring the 21 athletes before and after the completion of a 3000-meter run on an outdoor track and field facility. Inclusion criteria required participants to be at least 18 years old and engage in running-related sports at least three times per week. Target pace was set at 4–5 min/km for men and 5–6 min/km for women. Heart rates were measured using wrist-worn (Polar Pacer Pro) monitors at two points: before the run and after 3000 meters. Immediately following each heart rate measurement, short video clips were recorded to capture the face and upper bodies (within seconds after crossing the finish line). After each video recording, participants self-reported their perceived exertion by marking an "X" on the Borg scale displayed in an Excel sheet on a laptop. Each participant was filmed two times: once at rest before the race, and once after 3000m (we again had overt and covert recordings of athletes, which again were rated very similarly; no difference in dependent t-test; $p = .88$). Hence, only overt recordings of athletes were used for further analyses. All video frames were standardized without sound to show only the face, shoulders, and upper chest to focus on nonverbal signals. Otherwise everything was identical to Study 1.

Results

ANOVA

The descriptive statistics are summarized in Fig. 1, middle panel. A 2 (own vs. observer rating) $\times$ 2 (before vs. after exertion) ANOVA revealed no main effect of rating type, $F(1, 20) = 0.09$, $p = .762$, $\eta^2_p = .00$, a significant main effect of exertion, $F(1, 20) = 479.53$, $p < .001$, $\eta^2_p = .996$, and a significant interaction, $F(1, 20) = 24.80$, $p < .001$, $\eta^2_p = .554$. As expected, exertion ratings increased substantially after the race. Simple effects again showed that exertion ratings were higher post-sprint than at baseline for both self-ratings and observer-ratings (both $p < .001$). Follow-up simple effects analyses further clarified the significant interaction: observers again underestimated post-race exertion relative to athletes' own ratings ($p = .016$), whereas in the rested condition, observers provided higher ratings than the athletes themselves ($p = .001$).

Correlational Analyses

Figure 2, middle panel illustrates the strong positive correlation between own and averaged observer RPE, $r(42) = .846$, $p < .001$. Figure 5 (top square) illustrates the strong positive correlations of athletes' own RPE scores with heart rate, $r(42) = .963$, $p < .001$. A similarly strong pattern—albeit smaller—emerged for aggregated observer RPE ratings and heart rate in the 3000-m adult track race, $r(42) = .880$, $p < .001$. To formally compare ownRPE–heart rate and observerRPE–heart rate correlations, we used the *cocor* package for overlapping dependent correlations (Diedenhofen & Musch, 2015), applying two-sided tests with $\alpha = .05$. Own-RPE correlations with heart rate were significantly higher than observer-RPE*heart rate correlations in the 3000-m adult track ($p < .005$).

Discussion

The results of Study 2 mirrored the high convergence found in Study 1, with strong correlations between athletes' RPE and aggregated observer ratings. While both athletes and observers tracked the massive increase in exertion from baseline to post-race, the interaction again highlighted significant differences in absolute agreement. Specifically, observers tended to overestimate exertion at baseline but systematically underestimated it following the race compared to athletes' self-reports. Furthermore, while both rating types correlated strongly with heart rate, athletes' own RPE demonstrated significantly higher physiological sensitivity than observer judgments. These findings suggest that while observers accurately capture the relative intensity of a 3,000-meter effort, they may be less attuned to the specific physiological peaks of internal strain than the athletes themselves.

To further test the generalizability of findings, Study 3 extended the logic of Study 1 and 2 to the context of an children 400-meter track race.

Study 3: Children 400m Track

Method

Participants

The observer sample estimated the exertion of 21 different track and field children athletes (9 boys and 12 girls), aged 6 to 8 years. All recruited athletes completed the study. The observer sample of 57 participants (38 female and 19 male) with an average age of 26.28 years ($SD = 8.03$). Only participants with complete data in the online survey were included. Sample size considerations were identical to Study 1. Informed consent was obtained from every child, parent, and observer participant. Ethical approval for research on minors was granted by the Ethics board of the German Sport University Cologne. Prior to data collection, a comprehensive information sheet detailing the study's procedures, risks, and benefits was distributed to the parents or legal guardians of all participating child athletes. Written informed consent was obtained from the parents before any involvement in the study. Additionally, the purpose of the research was explained to the athletes in age-appropriate language, and verbal or written assent was secured from each child to ensure their voluntary participation. All participants and their guardians were informed of their right to withdraw from the study at any point without penalty.

Procedure and Stimuli Creation

The procedure and video recordings were highly similar to the previous studies and therefore only the important differences are described here. For Study 3, children were instructed to complete a 400-meter run at a self-determined brisk pace. Heart rate (Polar Pacer Pro) and perceived exertion were recorded before and after the run. Short video clips were captured at the beginning and end of each run on an outdoor track and field facility (within seconds after crossing the finish line). This stimulus set consisted of 42 video clips, each lasting between two and five seconds. A total of 21 children (9 boys and 12 girls), aged 6 to 8 years, com-

pleted the 400-meter run. Most participants were members of an athletics club, though their training levels and experience varied. Before the run, the Borg RPE scale was thoroughly explained to the children, supported by elements from the Facial Rating of Perceived Exertion (FRPE) scale (Chen et al., 2017). To enhance comprehension, each numerical value on the Borg scale was paired with a corresponding facial expression ('smiley faces'). After the children provided their subjective exertion ratings, heart rate was measured using a fingertip pulse oximeter immediately before the run. Video recordings were taken at the start and end of the 400-meter run. Upon reaching the finish line, heart rate was measured again without delay, and subjective exhaustion was reassessed using both the Borg and FRPE scales. All videos were recorded without audio to ensure evaluations were based solely on visual cues.

Results

ANOVA

The descriptive statistics are summarized in Fig. 1, right panel. A 2 (own vs. observer rating) $\times$ 2 (before vs. after exertion) ANOVA revealed a significant main effect of rating type, $F(1, 20) = 6.71, p = .018, \eta^2_p = .251$, a significant main effect of exertion, $F(1, 20) = 120.31, p < .001, \eta^2_p = .857$, and a significant interaction, $F(1, 20) = 18.06, p < .001, \eta^2_p = .475$. As expected, exertion ratings were higher after the run (both own and other ratings were lower at baseline than after the race, $p < .001$). Post-hoc simple effect analyses: the interaction shows that observers only slightly—and nonsignificantly—underestimated post-race exertion relative to children's own ratings ($p = .567$), whereas in the rested condition, observers provided higher exertion ratings than the children themselves ($p = .001$).

Correlational Analyses

Figure 2, right panel illustrates the strong positive correlation between own and averaged observer RPE, $r(42) = .813, p < .001$. Figure 5 (bottom square) illustrates the strong positive correlations of children athletes' own RPE scores with heart rate, $r(42) = .886, p < .001$. A similarly strong pattern—albeit smaller—emerged for aggregated observer RPE ratings and heart rate, $r(42) = .795, p < .001$. Again, own-RPE correlations with heart rate were significantly higher than observer-RPE*heart rate correlations in the 400-m children track race ($p < .05$).

Discussion

Study 3 further confirmed the strong convergence between self-reports and observer judgments, as reflected by the high correlation between children's RPE and aggregated thin-slice ratings. Consistent with previous studies, a significant interaction emerged: while observers provided higher exertion ratings than children at baseline, they reached a closer absolute agreement post-race, showing only a slight, non-significant underestimation of intensity. Although both rating types correlated strongly with heart rate, children's self-reports again demonstrated significantly higher physiological sensitivity than observer judgments. Collectively, these results suggest that while observers accurately detect the relative magnitude

of children's physical effort, their judgments remain less precisely aligned with internal physiological markers than the athletes' own perceptions.

To further test the generalizability of findings, Study 4 extended the logic of Study 1–3 to the context of an adult home trainer cycling workout.

Study 4: Adults Cycling

Method

Participants

The observer sample estimated the exertion of 20 different cyclists (10 female and 10 male), aged 25 on average. All recruited athletes completed the study. The observer sample of 55 participants (31 female and 24 male) with an average age of 26.12 years ($SD = 3.44$). Only participants with complete data in the online survey were included. Sample size considerations were identical to Study 1. Informed consent was obtained from every athlete and observer participant.

Procedure and Stimuli Creation

The procedure and video recordings were highly similar to the previous studies and therefore only the important differences are described here. For Study 4, the created stimulus set comprised video sequences of cyclists recorded before, after moderate, and after high-intensity cycling sessions on an indoor home trainer. For the recordings participants got off their bikes and filming started within seconds after the exertion. Heart rate was measured via pulse oximeter (Polar Pacer Pro). All participants completed an FTP test (Functional Threshold Power) beforehand to determine individual performance metrics. The FTP represents the maximum power a person can sustain for one hour of cycling (Allen & Coggan, 2006). This enabled all participants to complete cycling sessions of equivalent intensity on a home trainer, based on their FTP. The stimuli comprised 60 video clips, each lasting four to six seconds, featuring participants recorded three times: before exertion, after moderate exertion, and after intense exertion. Participants were informed in advance about the recordings and provided informed consent. They were also briefed on the Borg scale, which they later used for self-assessment. All videos were recorded without audio to ensure evaluations were based solely on visual cues. The footage consistently showed only the upper body of participants seated upright on the bicycle, recorded with the same camera against a neutral background. Uniform lighting conditions and similar clothing (black tops) were maintained throughout the study to minimize external influences on the raters. After evaluating all video sequences in the cycling study, participants were asked to enter into a free-text field those indicators of exertion that they considered most important for their exertion ratings.

Results

ANOVA

The descriptive statistics are summarized in Fig. 3, left panel. A 2 (own vs. observer rating) $\times$ 3 (before vs. mid vs. after exertion) ANOVA showed a significant main effect of exertion, $F(2, 38) = 472.45$, $p < .001$, $\eta^2_p = .961$, no main effect of rating type, $F(1, 19) = 2.22$, $p = .152$, $\eta^2_p = .105$, and a significant interaction, $F(2, 38) = 11.34$, $p < .001$, $\eta^2_p = .374$. As expected, exertion ratings increased linearly with cycling load. Simple effect analyses revealed significant differences between all three measurement points within both the own and other ratings (all $p < .001$). Follow up simple effect analyses to illuminate the interaction indicated that observers underestimated exertion and high load ($p = .008$) relative to athletes' own RPEs (not significant at mid load, $p = .242$), whereas in the rested condition observers provided higher ratings than the athletes ($p = .002$).

Correlational Analyses

Figure 4, left panel illustrates the strong positive correlation between own and averaged observer RPE, $r(60) = .828$, $p < .001$. Figure 5 (second square) illustrates the strong positive correlations of athletes' own RPE scores with heart rate, $r(60) = .910$, $p < .001$. A similarly strong pattern—albeit smaller—emerged for aggregated observer RPE ratings and heart rate, $r(60) = .783$, $p < .001$. Again, own-RPE correlations with heart rate were significantly higher than observer-RPE*heart rate correlations ($p < .005$).

Discussion

Study 4 demonstrated a high degree of convergence between self-reports and observer judgments during a cycling task, characterized by a strong correlation between athletes' RPE and aggregated ratings. While both athletes and observers tracked the linear increase in exertion across all loads, a significant interaction revealed that observers again provided higher ratings than athletes at baseline but underestimated intensity at peak load. Furthermore, while both rating types were strongly associated with heart rate, athletes' self-reports showed significantly greater physiological sensitivity than observer judgments. These findings rein-

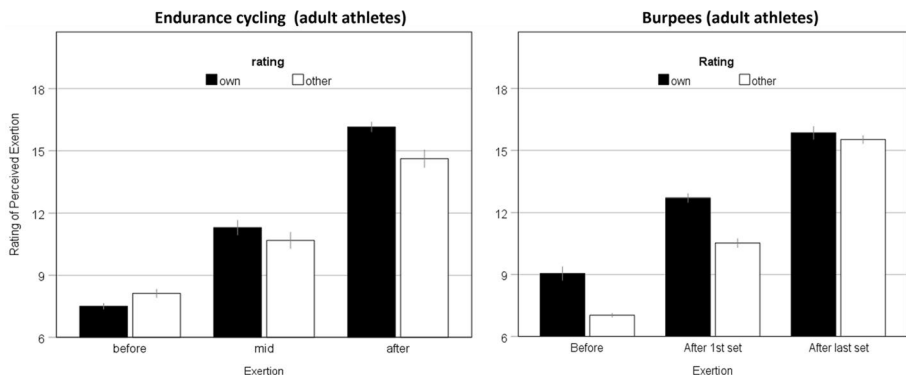


Fig. 3 Summary of descriptive results from Studies 4 and 5. Error bars represent standard errors

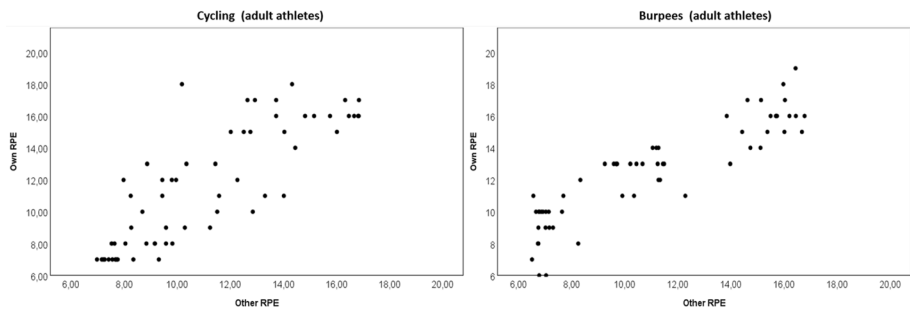


Fig. 4 Scatter plots illustrating the relationship between own ratings and observer ratings of RPE for Study 4 and 5

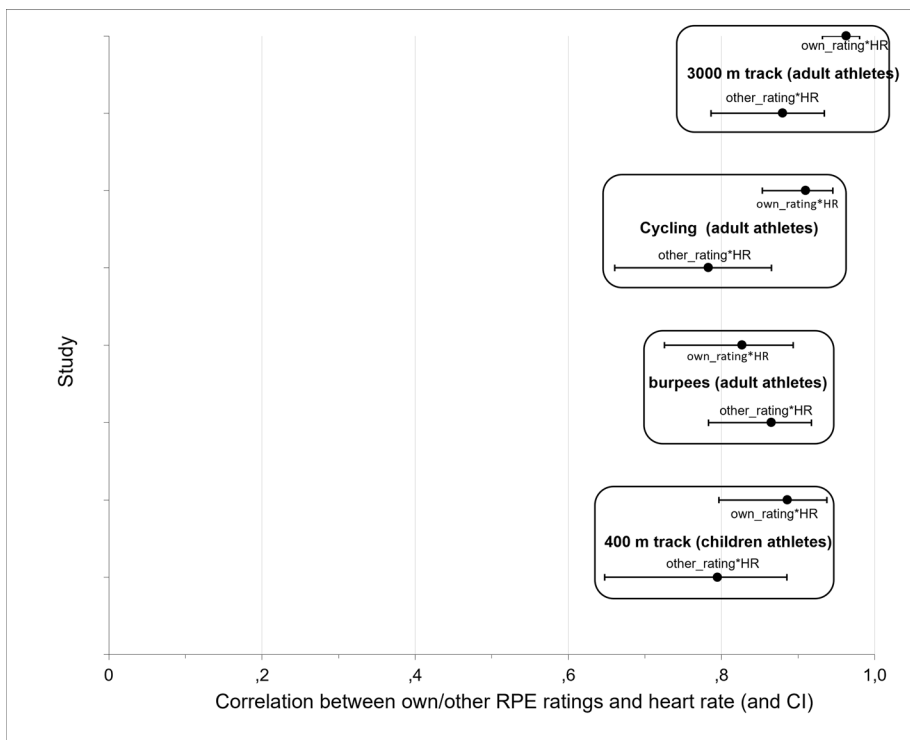


Fig. 5 Pearson correlation coefficients and 95% CI for the own RPE*heartrate and the other RPE*heartrate data sets

force the pattern that while external observers accurately detect relative shifts in effort, their sensitivity to peak internal strain remains lower than that of the athletes.

To further test the generalizability of findings, Study 5 extended the logic of Study 1-4 to the context of an adult gym/burpee workout.

Study 5: Adults Workout/"Burpees"

Method

Participants

The observer sample estimated the exertion of 20 recreational workout athletes (11 female and 9 male), aged 23 on average. All recruited athletes completed the study. The observer sample of 54 participants (28 female and 26 male) with an average age of 27.87 years ($SD = 5.34$). Only participants with complete data in the online survey were included. Sample size considerations were identical to Study 1. Informed consent was obtained from every athlete and observer participant.

Procedure and Stimuli Creation

The procedure and video recordings were highly similar to the previous studies and therefore only the important differences are described here. For Study 5, the stimuli comprised 60 sequences, with three clips per athlete. The twenty athletes performed four sets of ten burpees each in an indoor gym. The exercise type involved the performance of burpees involving four distinct movements. The exercise begun in a standing position, which was followed by a supported squat with both hands touching the ground. The third movement involved the participant kicking their feet back and touching the ground to form a plank, and lastly the participant returned to a standing position with their arms extended above the head, concluding with a clap. This motion would be repeated by the participant until the completion of ten burpees. Videos were sampled before the burpees, after the first set of ten burpees, and after the fourth set of 10 burpees. Participants were filmed for approximately five seconds before exercise and immediately (within seconds) after each set. During the 45-second rest intervals, participants discreetly rated their exertion using the Borg scale and had their heart rates measured via pulse oximeter (Polar Pacer Pro). Each video, about five seconds long, captured the face and upper body.

Results

ANOVA

The descriptive statistics are summarized in Fig. 3, right panel. A 2 (own vs. observer rating) $\times$ 3 (before vs. mid vs. after exertion) ANOVA revealed a significant main effect of exertion, $F(2, 38) = 549.98, p < .001, \eta^2_p = .967$, a significant main effect of rating type, $F(1, 19) = 43.78, p < .001, \eta^2_p = .697$, and a significant interaction, $F(2, 38) = 16.83, p < .001, \eta^2_p = .470$. Exertion ratings increased linearly with training load (Simple effect analyses revealed significant differences between all three measurement points within both the own and other ratings, all $p < .001$), and overall self-ratings were higher than observer ratings. Simple effect analyses on the interaction indicated that observers underestimated exertion at low ($p = .001$) and mid load ($p = .001$), whereas at high load observer and self-ratings did not differ ($p = .275$).

Correlational Analyses

Figure 4, right panel illustrates the strong positive correlation between own and averaged observer RPE, $r(60) = .899, p < .001$. Figure 5 (third square) illustrates the strong positive correlations of athletes' own RPE scores with heart rate, $r(60) = .827, p < .001$. A similarly strong pattern emerged for aggregated observer RPE ratings and heart rate, $r(60) = .865, p < .001$. This time, own-RPE correlations with heart rate were comparable to observer-RPE*heart rate correlations ($p = .019$).

Discussion

Study 5 replicated the high convergence between self-reports and observer judgments, evidenced by a strong correlation between athletes' RPE and aggregated ratings. While both athletes and observers tracked the linear increase in exertion across training loads, a distinct interaction emerged: unlike previous studies, observers' judgments aligned closely with self-reports at high intensity but significantly underestimated exertion at low and mid loads. Notably, in this study, the physiological sensitivity of observer ratings was comparable to that of self-reports, as both correlated strongly and similarly with heart rate.

General Discussion

The present research investigated whether observers can accurately detect physical exertion from brief video clips and how observer ratings correspond to athletes' self-reported exertion and physiological markers. Across five studies, observers consistently tracked increases in exertion: aggregated observer ratings rose as physical load intensified in every dataset. Correlational analyses further supported this pattern, revealing strong positive associations between self- and observer ratings as well as between observer ratings and heart rate. These results indicate that observers extracted valid cues from brief recordings of both adults and children. Consistent with prior work (Liu et al., 2025), self-ratings tended to show slightly stronger associations with heart rate than observer ratings, suggesting marginally higher accuracy for interoceptive assessments. A notable applied finding was that observers systematically underestimated exertion at moderate and high levels of load, implying a potential risk of coaches unintentionally overloading athletes when relying solely on visual judgment.

Overall, the findings support the thin-slice paradigm and the retrodiction framework in the context of physical exercise. Humans appear well-equipped to infer exertion states from minimal behavioral information, reinforcing the broader notion that people can form accurate judgments based on brief expressive cues (Ambady et al., 2000; Bogdan, 1997). Darwin (1871) similarly argued that humans evolved a sensitivity to signals revealing others' internal states, including fatigue. In line with this evolutionary perspective, observers in the current studies reliably detected whether athletes had exerted themselves. This pattern also resonates with Darwin's (1872) anti-creationist argument that nonverbal expressions are not purposefully designed; showing fatigue may be disadvantageous to the expresser. The results correspond with the Behavioral Ecology Theory (Crivelli & Fridlund, 2018; Fridlund, 1994), which conceptualizes nonverbal behavior as socially functional signals

that guide interaction. Future research should examine how exertion-related cues influence subsequent social behaviors and decisions.

The high observer accuracy aligns with prior interpersonal accuracy literature (Ambady & Rosenthal, 1992; Carney et al., 2007) and can partly be understood through “wisdom of the crowd” processes (Furley et al., 2025; Surowiecki, 2004). Aggregating diverse judgments reduces noise and enhances accuracy, but this does not imply the effects are mere statistical artefacts. Aggregation works precisely because expressive behavior contains meaningful—albeit probabilistic—information. Single expressive instances are noisy, but consistent patterns emerge across aggregated thin slices, allowing reliable inference. A potential limitation of the present work involves the comparison of aggregated observer ratings from over 50 individuals against the single interoceptive rating of the athlete. While this approach is standard in thin-slices research—such as when comparing external observations of personality with self-reports (Murphy & Hall, 2021)—the conceptual commensurability of these metrics remains a point of scholarly debate. However, this asymmetry is inherent to the research paradigm, as interoceptive signals of exertion are uniquely accessible to the athlete. By integrating these findings with objective heart-rate data, the present work provides converging evidence across five studies that individual RPE aligns closely with both physiological markers and external observer judgments.

From a theoretical standpoint, the findings relate back to Borg’s (1962) conception of perceived exertion as a “Gestalt” composed of multiple sensory and contextual factors. Interestingly, this Gestalt appears accessible through both interoception (Craig, 2002) and external observation. Across various modalities—including 200-m sprints (Study 1), 3000-m runs (Study 2), stationary cycling (Study 4), and high-intensity burpees (Study 5)—observers consistently rated exertion as higher than baseline. However, a systematic bias emerged at higher intensities, where observers significantly underestimated adult athletes’ self-reported RPE. Notably, this discrepancy was absent in the children sample (Study 3), where self-ratings and observer judgments remained congruent following exertion. This systematic underestimation in adults warrants further investigation, as it suggests that practitioners may unintentionally prescribe loads that exceed safe physiological limits. Furthermore, the lack of divergence between observer and self-ratings in the pediatric sample (Study 3) may initially suggest developmental limitations in interoceptive accuracy among 6- to 8-year-olds (Gros Lambert & Mahon, 2006). However, the robust correlation observed between the children’s self-reported RPE and their heart rate ($r = .81$) contradicts the notion of poor self-perception. Instead, this suggests that young children may express exertion through nonverbal cues that differ from those of adults, a possibility that warrants further empirical investigation.

A limitation of the present work is that we did not systematically quantify visible cues that varied with exertion, as previous research has done (De Morree & Marcora, 2010, 2012; Uchida et al., 2018). Nonetheless, exploratory self-reports from the cycling study provide initial insight into which cues observers rely on. Participants most frequently cited breathing (87%), skin color (84%), and perspiration (66%) as diagnostic cues, followed by posture (26%), facial expression (24%), and gaze behavior (9%). These categories could serve as a basis for future lens-model analyses (Brunswick, 1956) to determine cue validity and cue utilization in exertion perception.

A notable applied contribution is the extension of Borg’s RPE scale. Beyond its established use for self-assessment, the present findings suggest that the RPE scale can serve as a practical observational tool for coaches, therapists, and teachers. Given the high correspondence between self- and observer ratings, the scale may help practitioners estimate exertion without directly querying the athlete.

The research also has strengths and limitations. Strengths include the randomized selection and presentation of video clips, reducing order effects; replication across five studies, indicating robust effects; and the integration of physiological, self-report, and observer-based measures—an approach not previously used in exertion-perception studies. Limitations include modest sample sizes, although aligned with prior thin-slice research (Ambady & Rosenthal, 1992). A notable limitation of the present research is the lack of detailed demographic information regarding the race and ethnicity of the participants. While data were collected within a German context, specific information on cultural background or migration history was not recorded. Given that the expression and perception of nonverbal cues can vary significantly across cultural and racial contexts, the reliance on a potentially homogenous, Westernized sample may limit the generalizability of findings to more diverse populations. Future research should prioritize the inclusion of varied demographic metrics to determine whether the observed correspondence between interoceptive RPE and observer judgments holds across different cultural frameworks and nonverbal display rules.

In conclusion, this research provides the first systematic evidence that observers can reliably detect physical exertion in others from brief behavioral samples. Strong convergent correlations between self-reported exertion, observer ratings, and heart rate suggest that expressive behavior carries meaningful information about underlying physiological states. These findings offer important theoretical and applied implications for social perception, sport science, and exertion monitoring in everyday physical activity contexts.

Author Contribution - (single author submission)

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Data Availability All data can be found here: [<https://osf.io/nbxmd/overview>]

Declarations

Conflict of interest The authors declare no competing interests.

Research Disclosure Statements The authors declare that 1) (a) the total number of excluded observations and (b) the reasons for making these exclusions have been reported in this manuscript; 2) that all independent variables or manipulations, whether successful or failed, have been reported in the manuscript; 3) that all dependent variables or measures that were analyzed for this article's target research question have been reported in the manuscript/online supplements; 4) that (a) how sample size was determined and how sample sizes were determined has been reported in the manuscript.

Consent to Participate Informed consent was obtained online via a button press.

Permission to Reproduce Material from Other Sources Not applicable.

Ethics Approval The study was approved by the local university's ethics committee.

Open Science All data can be found here: https://osf.io/w4mky/overview?view_only=902e13789fb849f483800fca6923f27c

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